

Lecture 4: Determinants and Linear Systems

MATH 2308

February 23, 2023

When is a matrix invertible?

In the last lecture, we learned that the 2×2 matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible if and only if $ad - bc \neq 0$. We give this value a name.

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The 2×2 determinant

The **determinant** of the 2×2 matrix

$$\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

denoted $\det \mathbf{A}$ or $|\mathbf{A}|$, is $ad - bc$.

Examples

$$\begin{vmatrix} 2 & 4 \\ 3 & 1 \end{vmatrix}$$

$$\begin{vmatrix} 4 & 1 \\ 8 & 2 \end{vmatrix}$$

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$$\begin{vmatrix} 2 & 4 \\ 3 & 1 \end{vmatrix} = 2 \cdot 1 - 4 \cdot 3 = -10$$

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The matrix is not invertible. Recall that another name for a non-invertible matrix is **singular**.

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Also note that the second row is a multiple of the first.

Determinants of Larger Matrices

Larger square matrices also have determinants, though it can get complicated. Here's a trick for a 3×3 matrix. If

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

augment the matrix by adding copies of the first two columns on the right.

$$\left| \begin{array}{ccc|cc} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{array} \right|$$

3×3 determinants, continued

With

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix},$$

we have

$$\det \mathbf{A} = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{31}a_{22}a_{13} - a_{32}a_{23}a_{11} - a_{33}a_{21}a_{12}.$$

3×3 determinants, continued

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Downward diagonal products are added and upward diagonal products are subtracted.

3×3 example

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$$\begin{aligned} |\mathbf{A}| &= (1)(2)(0) + (2)(1)(1) + (3)(0)(3) - (1)(2)(3) - (3)(1)(1) - (0)(0)(2) \\ &= 0 + 2 + 0 - 6 - 3 - 0 = -7. \end{aligned}$$

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You can use technology to find determinants for larger square matrices.

Determinants and invertible matrices

What was true for 2×2 matrices is true for larger square matrices.

Theorem: Invertible matrices

An $n \times n$ matrix $\mathbf{A}$ is invertible if and only if $\det \mathbf{A} \neq 0$.

Determinants of products

The determinant of a product of square matrices is nice.

The determinant of a product

Let **A** and **B** be $n \times n$ matrices. Then

$$\det(\mathbf{AB}) = \det(\mathbf{BA}) = \det \mathbf{A} \det \mathbf{B}.$$

Diagonal and triangular matrices

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$$\mathbf{A} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

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$$\det \mathbf{A} = 2 \cdot (-1) \cdot 4 = -8.$$

Upper and lower triangular matrices

A matrix is called **upper triangular** if all of the entries *below* the main diagonal are zero. A matrix is **lower triangular** if all of the entries above the main diagonal are zero. For example

$$\mathbf{U} = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 5 & 1 \\ 0 & 0 & 2 \end{bmatrix} \text{ and } \mathbf{L} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ 4 & 5 & 6 \end{bmatrix}$$

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The determinant of a triangular matrix is also the product of the diagonal entries, so

$$\det \mathbf{U} = 20 \text{ and } \det \mathbf{L} = -6.$$

Linear systems as a matrix product

Consider the product

$$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

=

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The entries in $\mathbf{Ax}$ make up the left hand side of a system of equations! We can write a system of linear equations as a matrix equation.

Systems of equations as matrix equations

The system of equations

$$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1$$

$$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2$$

$$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3$$

can be written

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

or

$$\mathbf{Ax} = \mathbf{b}.$$

Solving systems using matrix inverses

If A is an invertible $n \times n$ matrix, then

$$\begin{aligned}\mathbf{A}^{-1}(\mathbf{Ax}) &= \mathbf{A}^{-1}\mathbf{b} \\ \mathbf{x} &= \mathbf{A}^{-1}\mathbf{b}\end{aligned}$$

gives us our solution.

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Theorem

Let $\mathbf{A}$ be an invertible $n \times n$ matrix. Then the matrix equation

$$\mathbf{Ax} = \mathbf{b}$$

has the unique solution

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}.$$

Example

Solve

$$\begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

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If $\mathbf{A} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$, then

$$\mathbf{A}^{-1} = \frac{1}{3 \cdot 1 - 1 \cdot 2} \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix},$$

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so

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

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- Computationally speaking, it's more efficient to solve a system of equations using row reduction than by taking inverses.
- If $\mathbf{A}$ is singular, there still can be a solution (actually, infinitely many). Row reduction will always work.